# Chapter 20
# A Critically Endangered Capuchin Monkey (*Sapajus xanthosternos*) Living in a Highly Fragmented Hotspot

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**Abstract** Capuchin monkeys (*Sapajus* sp.) have cognitive skills and variation in morphological traits among different species that allow for a varied diet and flexible use of food resources. In the northern Atlantic forest, capuchins have been forced by fragmentation to live in human-altered environments; in addition, animal densities have been reduced by hunting and habitat loss, wiping out a number of large mammals that disperse zoochoric fruits. With the decrease in populations of other seed dispersers, yellow-breasted capuchins (*Sapajus xanthosternos*) are currently one of the largest fruit-eating mammals in the region. We provide frequency of feeding on invasive species (*sensu* Colautti and MacIsaac, Divers Distrib 10(2):135–141, 2004), and insights into how they might compete for seed dispersal with other species. *S. xanthosternos* had a high frequency of travelling and feeding on fruits. They ate fruits in primary, old- and young-secondary forest, swamp, and *cabruca* (shaded cocoa plantations). In addition, they fed on fruits of oil-palm (*Elaeis guineensis*), jackfruit (*Artocarpus heterophyllus*) and cocoa (*Theobroma cacao*); these cultivars compete with native species for seed dispersal by *S. xanthosternos*. We recommend the management of invasive fruiting species, especially in protected areas.

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## Introduction

The Atlantic Forest of Brazil faces one of the higher extinction rates as compared to other biomes, due to its high species endemism, heavy historical and on-going habitat loss, increasing human population density (Laurance 2009), and increasing secondary forests in different levels of regeneration. The entire biome is threatened by deforestation and fragmentation with 83.4 % of forest fragments smaller than 50 ha. The northern Atlantic Forest is in a worse state of conservation than the southern part where the coastal areas of Bahia and Sergipe have less than 17 % of forest remaining (Ribeiro et al. 2009).

The yellow-breasted capuchin monkey (*Sapajus xanthosternos*) is endemic to the northern part of the Atlantic forest. Nowadays, remaining populations are extremely rare and restricted to small and degraded areas. *S. xanthosternos* is in critical danger of extinction due to habitat loss and hunting (IUCN 2010). The species has already disappeared over the major part of its original distribution (Kierulff et al. 2005). Thus, areas with the largest patches of forest and the largest number of capuchin groups are of great importance. Understanding how *S. xanthosternos* is coping with habitat disturbance is fundamental to future management of wild populations. Estimates of home range and habitat use are needed to establish the size of a suitable protection area and to determine management actions, such as connecting isolated fragments or expanding areas for the protection of a minimum viable population (Shaffer 1987).

Most of the large forest fragments harbouring populations of yellow-breasted capuchin monkeys are concentrated in southern Bahia Atlantic Forest (Kierulff et al. 2005). The presence of cocoa plantations in the agro-forestry system, with cocoa trees planted in forest understorey (called *cabrucas*), have helped to maintain a more forested landscape to preserve fauna and flora and to preclude a more dramatic change in microclimate and hydrological cycles (Mori et al. 1983), especially when compared to landscapes surrounded by pasture and extensive monocultures.

Cocoa (*Theobroma cacao*) was introduced in Bahia in 1746 from the Amazon and is still one of the most important commodities in southern Bahia Atlantic Forest (Purdy and Schmidt 1996). In the late 1980s, an economic crisis reached the region when a fungus, named witch's broom (*Moniliophthora perniciosa*), killed a vast area of cocoa plantations (Purdy and Schmidt 1996). This resulted in the conversion of old agro-forestry into pasture and other less environmental-friendly crops (e.g. manihot), and in the degradation of many of the forest remnants (Alger and Caldas 1994; Saatchi et al. 2001). Other exotic plants ubiquitous in Bahia Atlantic forest are oil-palm (*Elaeis guineensis*) that was introduced in Bahia in the late 1500s (Bastos et al. 2001), and abundantly used in Bahian cuisine, and jackfruit (*Artocarpus heterophyllus*), brought to Bahia for the first time in 1648 as an ornamental species (Ferrão 1993; Boni et al. 2009).

The classification of species as *exotic*, *invasive*, *introduced*, or *non-indigenous* has been currently discussed among invasion biologists and it is still subject to different views (Colautti and MacIsaac 2004; Richardson et al. 2000; Davis et al. 2001).

Here we consider cocoa, oil-palm and jackfruit to be invasive species in southern Bahia Atlantic forest, as all of them were introduced centuries ago from a different biome by humans, and today have a widespread distribution; also, they currently produce seedlings and might disperse longer distances if no management (removal and replacement by native species) is done. We follow a biogeographic approach and not taxonomic, i.e. the species per se is not invasive, but high dispersal capability and current distribution indicates their invasiveness in southern Bahia Atlantic Forest (Colautti and MacIsaac 2004; Richardson et al. 2000).

*Sapajus* are frugivore–insectivores (Robinson and Janson 1987) or faunifrugivores (Chivers 1998). Fruits, seeds, foliage (bromeliads), nectar and animal prey (including insects, birds, eggs, reptiles and small mammals) comprise their diet (Janson and Boinski 1992). Capuchin monkeys are able to feed on a wide range of food resources, not only due to their morphological and anatomical characteristics, but also because of their great manipulative and cognitive skills (Moura and Lee 2004; Canale et al. 2009). Crop use is common in environments with highly seasonal resources for primates (Freese and Oppenheimer 1981; Naughton-Treves et al. 1998; Di Bitetti 2001), but invasive plant species might be a threat to native species that also compete for seed dispersers (Gosper et al. 2005).

## Methods

### *Study Site*

Una Biological Reserve comprises one of the largest forest fragments in northern Atlantic forest, summing 18,500 ha (Fig. 20.1) It lies in southern Bahia, Brazil (15°06′–12′S, 39°02′–12′W) and its vegetation is composed by ca. 7,060 ha of mature and regenerating forests, broadly characterised as lowland moist tropical forest (Mori et al. 1983). Temperature averages 24–25 °C annually and rainfall is ca. 2,000 mm/year with no evident seasonality (Mori et al. 1983).

### *Behavioural Data Collection*

We observed behaviour and ranging patterns of 14 individuals (two adult males, three adult females, two sub-adult males, two sub-adult females, two juvenile males, two juvenile females, and one infant) in one group of *S. xanthosternos* from September 2007 to December 2008 (Fig. 20.2a, b). The group was radio-tracked (Kierulff et al. 2005) and trapped individuals were dye-marked, but most of the animals of the group were recognised by coat patterns, and sex and age could be determined in most cases during behavioural observations. By the time we started the study, the capuchin monkeys were completely habituated to the observers.

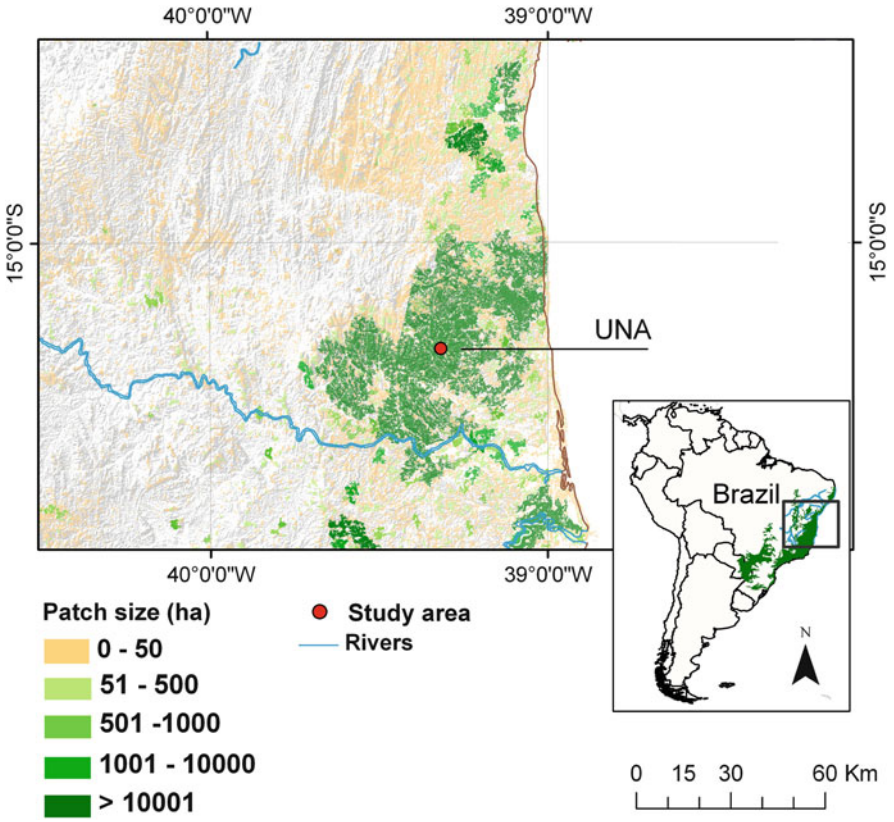



**Fig. 20.1** Largest forest fragment in the Atlantic Forest of southern Bahia where is located Una Biological Reserve (UNA)

Behavioural data were collected during 3–5 days each month using scan samples at 15-min intervals (Martin and Bateson 2007). Groups were followed most of the time from dawn to dusk ($N=55$ full days); when we were not able to follow the groups the whole day, we completed the period not sampled in the following day ($N=42$ half days). We collected data on behaviour in a total of 2,598 scans. To avoid bias towards more conspicuous behaviours, we weighted each scan equally by dividing each observation by the total number of observations in a scan.

We scored the behaviours: travel, rest, social, feed and other. *Travel* involved horizontal and vertical movements. *Feeding* was recorded when animals were manipulating, chewing or swallowing dietary items (fruit, flowers, sap/gum, pith, leaves, bromeliad (foliar base), invertebrates, vertebrates, eggs). *Rest* was recorded if animals were lying or simply not moving. *Social behaviours* included affiliative and aggressive interactions. *Other* included behaviours that could not be coupled and alone represented less than 1 % (chest-rubbing, drinking water, urine-washing, vigilance).

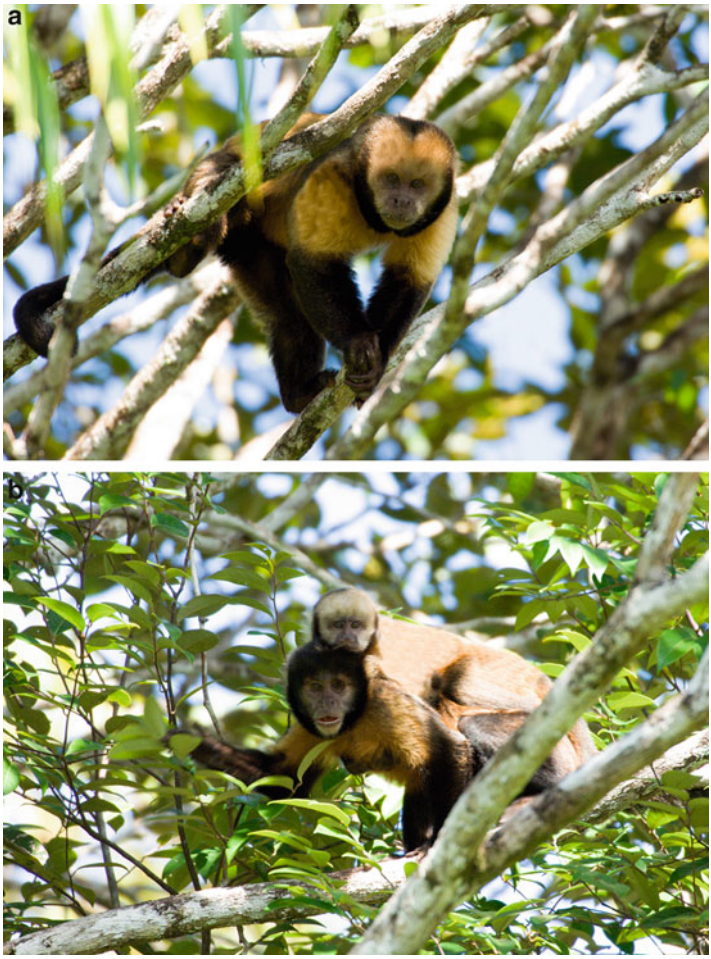



**Fig. 20.2** (**a**) *S. xanthosternos* adult male. Photo by Luciano Candisani. (**b**) *S. xanthosternos* adult female and infant. Photo by Luciano Candisani

## Results

### *Activity Budget*

Yellow-breasted capuchins' home range was 969 ha based on Minimum Convex Polygon. When calculated by kernel (95 %) home range was 671 ha. Core area (25 %) was 75 ha (Fig. 20.3).

Capuchins spent on average 32.8 % (±4.9 %) of their time feeding, frequencies range from 25.9 to 40.5 %. Most of the time animals were travelling or feeding, both behaviours together summed 75.9 % (±4.0 %), on average reaching 81.5 % of total behaviour frequency in 1 month (Oct 2007) (Fig. 20.4).

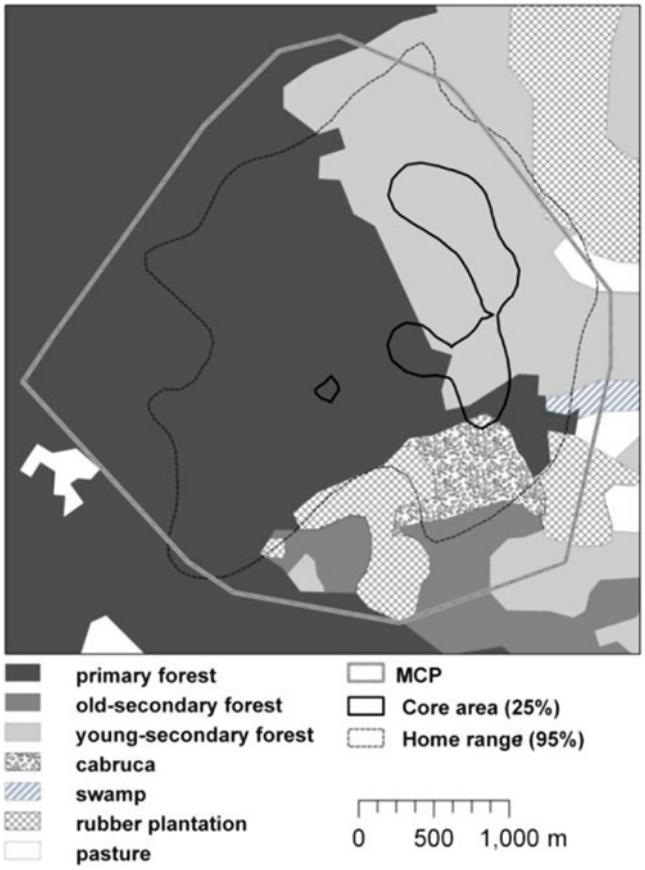



**Fig. 20.3** Home range of a capuchin monkey group at Una Biological Reserve based on data collected from September 2007 to December 2008

## *Food Resources Availability*

Fruits represent the bulk of *S. xathosternos*' diet. On average capuchins feed on fruits 38.3 % of total feeding frequency (±8.6 %, $N=14$ months) reaching up to 47.9 % of feeding frequency in February 2008. Invertebrates were the second most-frequent items in the diet, averaging 36.1 % (±6.6 %, $N=14$ months). In 4 months in 2008 (January, April, July and September) eating invertebrates was more frequent than eating fruits (Fig. 20.5), but there was no seasonal variation related to precipitation or fruit availability ($p>0.05$).

Capuchins feed on an average of 16.5 plant species/month. On average, 24.4 % (±15.6 %) of records of eating fruits were on invasive species (cocoa, jackfruit or oil-palm), but this went up to 47.1 % of fruit-feeding observations in 1 month (Oct 2007), and feeding on this three invasive species reached more than 25 % in 7 months ($N=14$ months) (Fig. 20.6).

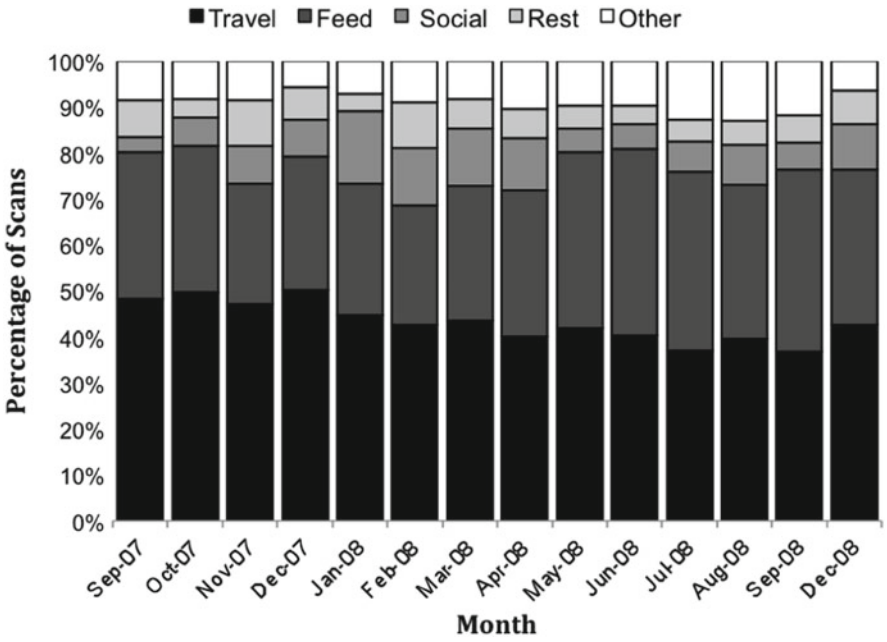



**Fig. 20.4** Activity budget for *S. xanthosternos* group at Una Biological Reserve from September 2007 to December 2008 ($N = 14$ months)

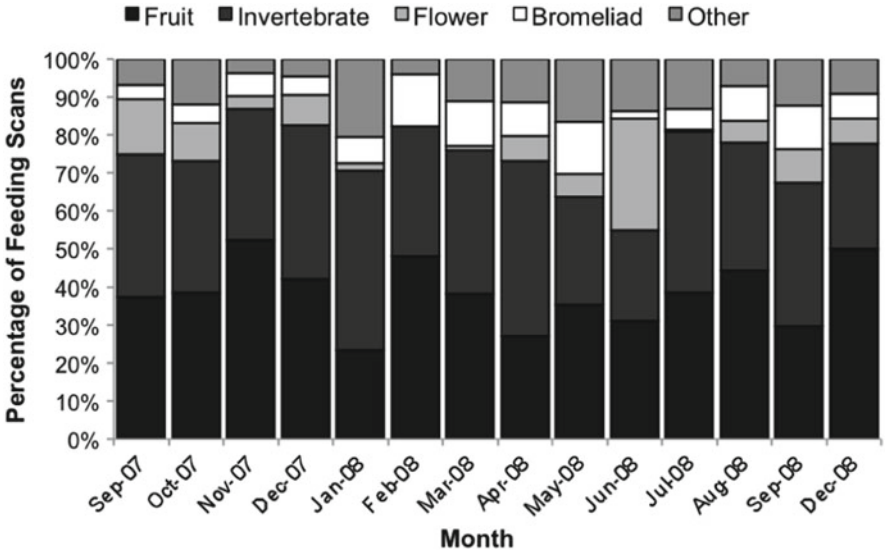



**Fig. 20.5** Percentage of feeding scans on different items for *S. xanthosternos* group at Una Biological Reserve from September 2007 to December 2008

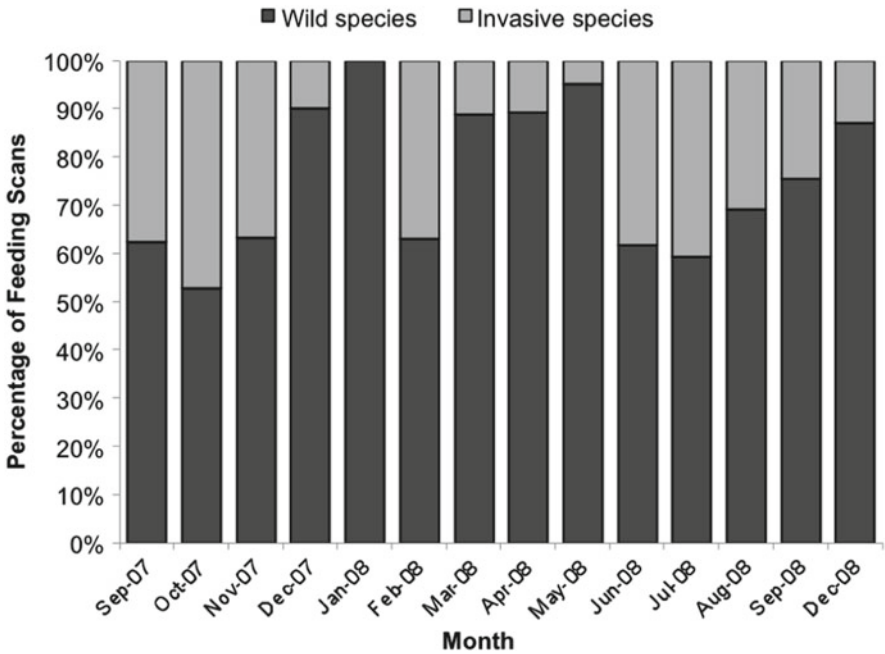



**Fig. 20.6** Percentage of scans that individuals were feeding on fruits of invasive species and wild species of trees for *S. xanthosternos* group at Una Biological Reserve from September 2007 to December 2008

**Table 20.1** Ten most-consumed plant species by *S. xanthosternos* in Una Biological Reserve from September 2007 to December 2008 ($N = 1{,}205$ feeding records)

| Family | Species | Feeding (%) | Status |
|---|---|---|---|
| Arecaceae | *Elaeis guineensis* | 26.1 | IN |
| Sapotaceae | *Manilkara longifolia* | 14.1 | ENA/EN |
| Menispermaceae | *Anomospermum reticulatum* | 7.7 | |
| Arecaceae | *Attalea funifera* | 6.6 | ENA |
| Mimosaceae | *Inga* sp. | 4.1 | |
| Sterculiaceae | *Theobroma cacao* | 3.6 | IN |
| Myrtaceae | sp. | 3.2 | |
| Moraceae | *Cecropia* sp. | 3.1 | |
| Melastomataceae | *Miconia prasina* | 2.6 | |
| | *Artocarpus heterophyllus* | 2.4 | IN |

*ENA* endemic of northern Atlantic forest, *EN* endangered, *IN* invasive species

From all feeding records ($N = 1{,}205$) oil-palm (*Elaeis guineensis*) was the most consumed item in their diet (26.1 %) and fruits of all invasive species summed 32.1 %. The second most-eaten fruit species was an endangered and endemic tree (*Manilkara longifolia*) (Table 20.1).

## Discussion

Primates have flexible behaviours that reflect environmental changes, providing predictive responses to ecological alterations that may guide conservation strategies. In Bahia Atlantic Forest large-bodied primates are in peril due to hunting and habitat loss: howler monkeys and woolly spider monkeys are virtually extinct in the region (Canale et al. 2012). Yellow-breasted capuchins are the next in line, as they are the largest primates to survive in that region, even though they are constantly hunted for bushmeat.

Yellow-breasted capuchins had one of the largest home ranges ever reported for a *Sapajus* species (969 ha). This is an important result for conservation action for this critically endangered species, since decision-makers aiming to establish reserves and protected areas will have to consider *S. xanthosternos*' need for a large forest to support a viable population.

Capuchin monkeys spend a lot of time foraging in different vegetation types and at different heights (Zhang 1995; Siemers 2000). This is especially the case for those living in fragmented and/or degraded forests. Yellow-breasted capuchin monkeys visited five different vegetation types within their home range: primary forest, old-secondary forest, young-secondary forest, swamp and *cabruca*. Travelling from one vegetation type to another might have an important effect on seed-dispersal patterns.

Seeds of cocoa and jackfruit were found in *S. xanthosternos* faeces still intact, but yellow-breasted capuchin monkeys were never observed swallowing seeds of oil-palm. In fact, cocoa seeds (3.0 ± 0.09 cm) (Chandel et al. 1995) and jackfruit seeds (3.1 ± 0.2 cm) (Lazure et al. 2010) are on the upper-limit size of seeds ingested by *Cebus* (<3 cm, Wehncke et al. 2003), which might not be different from *Sapajus* due to their morphological similarities. It is possible they could be dispersing invasive species around a protected area. We have not followed the germination of cocoa and jackfruit seeds defaecated by *S. xanthosternos*, but together with oil-palm they are already a concern for seed dispersal of native fruit species, as these three invasive species are competing for one of the most important seed dispersers remaining in Bahia Atlantic forest.

Although invasive species should not be kept inside protected areas, removing them could affect native frugivores relying on these resources (Gosper and Vivian-Smith 2009). Other endangered and endemic species might be depending on invasive plants on our study area. For instance, golden-headed lion tamarins (*Leontopithecus chrysomelas*) eat jackfruit and oil-palm fruits (Raboy and Dietz 2004), Wied's black-tufted-ear marmosets (*Callithrix kuhlii*) feed on jackfruit (Raboy et al. 2008), and a reintroduced population of red-billed currasows (*Crax bluemanbachii*) were observed feeding on oil-palm in another site in the southern Atlantic Forest (Bernardo et al. 2011). We observed collared peccaries (*Pecari tajacu*) and coatis (*Nasua nasua*) frequently eating those fruits as well.

With the replacement of forested areas by human-modified landscapes, primate populations are getting closer to humans and crops are becoming part of primate diets

all over tropical regions (Naughton-Treves et al. 1998; Chiarello 2003). Increase of crop-raiding is reported where there is a lack of wild fleshy-fruit species (de Freitas et al. 2008; Ludwig et al. 2006; Di Bitetti 2001). In this situation, crops could represent a fall-back food for primates (*sensu* Marshall and Wrangham 2007) as the use of non-indigenous species was related to low availability of wild fruits. Primates approaching crops only in times of low availability of wild food resources might not be related to low quality of crop foods compared to wild foods. In fact, most human crops are planted based on nutritional values, as much as edibility and palatability.

Feeding on crops is not always related to shortage of wild fruit species, as in some cases seasonal availability of wild fruits and crops coincides (Rímoli et al. 2008). In this study, there was no correlation between wild fruit availability and feeding on invasive species, probably because all crops were located inside forest fragments and mixed with wild tree species, reducing risk of predation by birds of prey. Even those crops close to humans and domestic animals were accessible through the canopy and mixed in dense wild vegetation. Alarm calls were given during human or dog approaches and the monkeys moved away quickly, back to forest patches far from human settlements. In fact, during interviews with local people harvesting cocoa, presence of capuchin monkeys was never reported in *cabrucas*, probably because capuchins retreat to the forest before being detected by humans, as observed by researchers following the capuchin group in this study.

The increase of crop-raiding during periods of wild fruit scarcity must be related to a trade-off between predation-risk and starvation. Crop-raiding must represent a high risk for primates, as they might be shot or harmed by humans and more easily predated due to reduced canopy cover (Naughton-Treves et al. 1998; Ludwig et al. 2006); even when crop-raiding in *cabrucas*, where canopy cover is mostly kept intact, capuchin monkeys are threatened by the constant presence of armed humans and dogs. During the lean season, however, primates must risk raiding crops to gain quality food. An extra risk is taken for crop-raiding by moving on the ground (e.g. sugar cane, maize and cassava); in these cases the attacks of domestic dogs seem to be fatal for *Sapajus* spp. (*S. nigritus*: Ludwig et al. 2006; *S. xanthosternos*: Canale, unpublished data).

Invasive species are not fall back foods for yellow-breasted capuchins, as they eat them all year around independently of wild fruit availability. They are indeed an important part of their diet, consumed independently of wild fruit availability. Thus, the common understanding of filler fall-back foods as low-quality foods eaten during periods of food shortage, does not apply to cocoa, jackfruit and/or oil-palm fruits in the yellow-breasted capuchin diet, as all of them are high-quality foods, clumped and of easy access. High-quality fruits could be used as staple fall-back foods (Marshall and Wrangham 2007). This would imply that those resources are used more often when there is lack of other preferable resources, which is not the case.

There is no general pattern in which invasive plants are consistently more appealing to frugivores than native species (Buckley et al. 2006). Thus, reforestation of degraded areas with native species that yield abundant fleshy fruits for local frugivores might reduce invasive plant species success in attracting seed dispersers

(Gosper et al. 2006). Managing invasive plants has received little attention from park managers (D'Antonio and Meyerson 2002; Gosper and Vivian-Smith 2009).

All in all, the long-term survival of large populations of these critically endangered primates must be currently dependent on the accessibility of nutritious invasive species; this must also be true to the other five species of capuchins leaving in highly degraded landscapes in the Atlantic Forest and in the *Cerrado*. Thus, the removal of invasive plant species should be carefully planned to avoid affecting animal populations; as they might suffer from the immediate reduction of food, and also, from drastic changes in the spatial–temporal availability of resources, which might affect use of space, ranging patterns, group encounter rates and, ultimately, group social stability.

## Conclusions

*S. xanthosternos* feed all year around on invasive species, such as oil-palm (*Elaeis guineensis*), jackfruit (*Artocarpus heterophyllus*) and cocoa (*Theobroma cacao*), swallowing seeds of the latter two. These invasive species are competing for seed dispersal by yellow-breasted capuchins; moreover, seeds of jackfruit and cocoa are being brought to natural environments across crop boundaries and human settlements, posing a real threat to protected areas that have them nearby.

Although the removal of invasive species would seem to be a straight-forward solution, those trees were brought to Bahia Atlantic Forest more than 250 years ago and nowadays they are important sources of food for a large group of endangered animals, such as yellow-breasted capuchin monkey (*S. xanthosternos*), golden-headed lion tamarin (*Leontopithecus chrysomelas*), Wied's black-eared marmoset (*C. kuhlii*), red-billed curassow (*C. blumenbachi*) and many other seed dispersers, such as coati (*Nasua nasua*), and collared peccary (*Pecari tajacu*).

The presence of non-indigenous plant species inside protected areas is being seriously neglected by authorities and conservationists in the Atlantic Forest. Land-use is one of the main issues in this relentlessly fragmented and human-encroached environment. Its ordination and planning must consider not only cropland locations and natural environment protection, but also plant–animal interactions and indigenous and non-indigenous species associations. Gradual and carefully planned removal of exotic plant species with aggressive invasiveness capability should be considered for protected areas and their buffer zones.

Finally, yellow-breasted capuchins are living in a fragmented environment surrounded by cropland and other human-modified habitats, such as forest in distinct levels of degradation. Its ability to crop-raid and exploit secondary forest, normally altered by human activities, imposes a new peril to its survival. Crop-raiding may generate human–animal conflict, which leads rural producers to hunt crop-raiders, such as capuchins. In addition, capuchins in human-modified habitats, such as secondary forest, may be more prone to be hunted or attacked by domestic dogs.

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